

DEPARTMENT OF CSE / CSC / CSE(AI&ML) / AI&ML / CSE(DS)

QUESTION BANK

NAME OF THE COURSE: DATABASE MANAGEMENT SYSTEMS

COURSE CODE: 24CSPC32

NAME OF THE FACULTY:

YEAR: II

SEMESTER: I

SECTION:

SHORT ANSWER QUESTIONS

UNIT – II

S No	Question	BTL	CO	POs / PSOs
1	What is the relational model?	1	CO2	PO1,PO2,PO3,PO12
2	Define a relation in a database context.	1	CO2	PO1,PO2,PO3,PO12
3	What is a tuple?	1	CO2	PO1,PO2,PO3,PO12
4	What is a primary key?	1	CO2	PO1,PO2,PO3,PO12
5	What are integrity constraints?	2	CO2	PO1,PO2,PO3,PO12
6	Name two types of integrity constraints.	1	CO2	PO1,PO2,PO3,PO12
7	What is referential integrity?	2	CO2	PO1,PO2,PO3,PO12
8	How are integrity constraints enforced?	2	CO2	PO1,PO2,PO3,PO12
9	What is the purpose of a foreign key?	2	CO2	PO1,PO2,PO3,PO12
10	Define relational algebra.	1	CO2	PO1,PO2,PO3,PO12
11	List any three relational algebra operators.	1	CO2	PO1,PO2,PO3,PO12
12	What is a selection operation in relational algebra?	2	CO2	PO1,PO2,PO3,PO12
13	What does the projection operator do?	2	CO2	PO1,PO2,PO3,PO12
14	Explain the difference between union and intersection operators.	2	CO2	PO1,PO2,PO3,PO12
15	What is the difference between tuple and domain relational calculus?	2	CO2	PO1,PO2,PO3,PO12
16	What is an E-R diagram?	1	CO2	PO1,PO2,PO3,PO12
17	How is an E-R diagram converted into relations?	3	CO2	PO1,PO2,PO3,PO12
18	Define a view in a relational database.	1	CO2	PO1,PO2,PO3,PO12
19	Why are views used in databases?	2	CO2	PO1,PO2,PO3,PO12
20	What happens when a table is dropped?	1	CO2	PO1,PO2,PO3,PO12
21	What is the consequence of altering a table?	2	CO2	PO1,PO2,PO3,PO12
22	What is the use of the join operator?	2	CO2	PO1,PO2,PO3,PO12

23	Define Cartesian product in relational algebra.	1	CO2	PO1,PO2,PO3,PO12
24	What does the rename operator do in relational algebra?	2	CO2	PO1,PO2,PO3,PO12
25	What is domain relational calculus?	1	CO2	PO1,PO2,PO3,PO12
26	What is tuple relational calculus?	1	CO2	PO1,PO2,PO3,PO12
27	What is an atomic attribute?	1	CO2	PO1,PO2,PO3,PO12
28	What is a composite attribute?	1	CO2	PO1,PO2,PO3,PO12
29	Explain about SET operators?	2	CO2	PO1,PO2,PO3,PO12
30	Discuss about the expressive power of relational algebra and calculus?	2	CO2	PO1,PO2,PO3,PO12
31	Define relational database query?	1	CO2	PO1,PO2,PO3,PO12
32	Explain different types of query languages?	2	CO2	PO1,PO2,PO3,PO12
33	Discuss the basic form of SQL query?	2	CO2	PO1,PO2,PO3,PO12
34	Define Null values?	1	CO2	PO1,PO2,PO3,PO12
35	What is a trigger in the context of enforcing integrity?	1	CO2	PO1,PO2,PO3,PO12
36	How is data retrieval different in relational algebra vs calculus?	2	CO2	PO1,PO2,PO3,PO12
37	Define tuple variable with its syntax?	1	CO2	PO1,PO2,PO3,PO12
38	What does the difference operator do?	2	CO2	PO1,PO2,PO3,PO12
39	What are null values in relational databases?	1	CO2	PO1,PO2,PO3,PO12
40	What is a domain in the relational model?	1	CO2	PO1,PO2,PO3,PO12

LONG ANSWER QUESTIONS

UNIT – II

S No	Question	BTL	CO	POs / PSOs
1	Explain the fundamental concepts of the relational model and its advantages over other data models.	2	CO2	PO1,PO2,PO3,PO12
2	Describe different types of integrity constraints and their importance in relational databases.	2	CO2	PO1,PO2,PO3,PO12
3	How are integrity constraints enforced practically in relational database systems?	3	CO2	PO1,PO2,PO3,PO12
4	Explain the steps involved in converting an E-R diagram into a relational schema.	3	CO2	PO1,PO2,PO3,PO12
5	Discuss the significance of primary keys, foreign keys, and candidate keys with examples.	2	CO2	PO1,PO2,PO3,PO12
6	Define relational algebra and describe its fundamental operators with examples.	2	CO2	PO1,PO2,PO3,PO12
7	Compare and contrast relational algebra and relational calculus in terms of expressiveness and usage.	4	CO2	PO1,PO2,PO3,PO12
8	Describe the tuple relational calculus and give examples of its syntax and use.	3	CO2	PO1,PO2,PO3,PO12
9	Explain domain relational calculus and how it differs from tuple relational calculus.	2	CO2	PO1,PO2,PO3,PO12
10	Discuss how views are created in relational databases and their practical uses.	3	CO2	PO1,PO2,PO3,PO12
11	What are the implications of dropping a table or view in a database? Discuss with examples.	2	CO2	PO1,PO2,PO3,PO12

12	Define join? Explain different types of joins?	1	CO2	PO1,PO2,PO3,PO12
13	Explain the relational algebra join operator and illustrate its types with examples.	2	CO2	PO1,PO2,PO3,PO12
14	Describe the selection and projection operators in relational algebra with real-world examples.	3	CO2	PO1,PO2,PO3,PO12
15	Discuss the union, intersection, and difference operators in relational algebra, highlighting their properties.	2	CO2	PO1,PO2,PO3,PO12
16	What are the challenges of enforcing integrity constraints in a multi-user environment?	4	CO2	PO1,PO2,PO3,PO12
17	Define Null value? Describe the effect of Null value in database?	1	CO2	PO1,PO2,PO3,PO12
18	Describe the process of logical database design starting from an E-R diagram to the final relational schema.	3	CO2	PO1,PO2,PO3,PO12
19	Discuss the difference between weak and strong entity sets in E-R modeling and their relational representation.	2	CO2	PO1,PO2,PO3,PO12
20	How do foreign keys enforce referential integrity? Illustrate with examples.	3	CO2	PO1,PO2,PO3,PO12
21	Illustrate different operations in relation algebra with an example?	3	CO2	PO1,PO2,PO3,PO12
22	Discuss different types of keys in relational databases with examples.	2	CO2	PO1,PO2,PO3,PO12
23	Illustrate with examples how the Cartesian product and join operations are related in relational algebra.	3	CO2	PO1,PO2,PO3,PO12
24	Explain how relational calculus queries are formed and interpreted by the database system.	2	CO2	PO1,PO2,PO3,PO12
25	Discuss the role of atomic and composite attributes in logical database design.	2	CO2	PO1,PO2,PO3,PO12
26	Convert a given ER diagram(with weak entity and 1:M relationship) into relation tables.	3	CO2	PO1,PO2,PO3,PO12
27	Explain the rename operator in relational algebra and its significance.	2	CO2	PO1,PO2,PO3,PO12
28	Compare tuple relational calculus and domain relational calculus in terms of syntax and semantics.	4	CO2	PO1,PO2,PO3,PO12
29	Discuss the importance of views for data security and abstraction in relational databases.	2	CO2	PO1,PO2,PO3,PO12
30	How does the database system maintain consistency when multiple users are updating the database?	4	CO2	PO1,PO2,PO3,PO12
31	Convert a given ER diagram with a many to many relationship and participation constraints.	3	CO2	PO1,PO2,PO3,PO12
32	Design an ER model for a university and convert it into a relational schema.	4	CO2	PO1,PO2,PO3,PO12
33	How do null values affect integrity constraints and query results in relational databases?	4	CO2	PO1,PO2,PO3,PO12
34	Describe the process and impact of dropping a view versus dropping a table.	2	CO2	PO1,PO2,PO3,PO12
35	Explain the importance of domain constraints and give examples of domain specification.	2	CO2	PO1,PO2,PO3,PO12
36	Discuss the impact of logical database design decisions on query performance.	4	CO2	PO1,PO2,PO3,PO12
37	Compare views and tables. Can all operations are performed on tables can be done on views?	4	CO2	PO1,PO2,PO3,PO12
38	Explain the concept of super key and how they relate to candidate keys and primary keys.	2	CO2	PO1,PO2,PO3,PO12
39	Compare the expressive power of relational algebra operators with SQL queries.	4	CO2	PO1,PO2,PO3,PO12
40	Describe a complex database query and show how it can be expressed in both relational algebra and calculus.	3	CO2	PO1,PO2,PO3,PO12